

Java Course Work Project

Smart Campus Student Management and Academic Information System

Technology: Java

Database: MySQL / SQLite

Recommended Framework: Java Spring Boot

Frontend: HTML, CSS, JavaScript, Bootstrap

Alternative Desktop Version: JavaFX

Development Environment: Visual Studio Code/IntelliJ IDEA / NetBeans / Eclipse

Database Access: JDBC / JPA-Hibernate

1. Project Title

Design and Development of a Smart Campus Student Management and Academic Information System Using Java

2. Background

Universities and institutions manage large amounts of student information, including registration, courses, examinations, attendance, fees, lecturers, departments and academic results.

When these activities are managed manually or through disconnected systems, institutions may experience:

- duplication of records;
- loss of information;
- delays in accessing student records;
- inaccurate examination results;
- difficulty tracking attendance;
- poor communication;
- inefficient report generation;
- unauthorized access to confidential information.

The proposed system will provide a centralized Java-based platform through which administrators, lecturers and students can manage and access academic information according to their roles.

3. Problem Statement

Many academic institutions still rely on manual procedures or multiple disconnected systems for managing student academic information.

For example, student registration may be performed separately from examination management, while attendance and fee information may be maintained in spreadsheets.

This creates difficulties in:

- maintaining accurate records;
- tracking student academic progress;
- generating reports;
- monitoring attendance;

- managing courses;
- publishing examination results;
- protecting student information.

Therefore, there is a need for an integrated system capable of managing major academic activities from a centralized platform.

4. Main Objective

To design and develop a Java-based **Smart Campus Student Management and Academic Information System** that automates student registration, course management, attendance, examination results, academic progression and reporting.

5. Specific Objectives

The system should:

1. Register and manage student information.
2. Manage academic programs and departments.
3. Manage courses and course units.
4. Assign lecturers to courses.
5. Register students for courses.
6. Record student attendance.
7. Record coursework and examination marks.
8. Automatically calculate student grades.
9. Generate student transcripts.
10. Monitor academic performance.
11. Generate academic reports.
12. Provide role-based authentication.
13. Maintain secure student records.
14. Provide dashboards for different users.

6. Target Users

The system will have three major users.

Administrator

The administrator manages:

- students;
- lecturers;
- departments;
- programs;
- courses;
- academic years;
- semesters;
- user accounts;
- system reports.

Lecturer

The lecturer can:

- view assigned courses;
- view registered students;
- record attendance;
- enter coursework marks;
- enter examination marks;
- view student performance;
- generate course reports.

Student

Students can:

- log into the system;
- view their profile;
- register for courses;
- view registered courses;
- view attendance;
- view results;
- view GPA;
- download/view academic transcript.

Deadline 25 Sept 2026, e campus using your GitHub Link